Course Title:
Basic Design (Introduction to Industrial Project)
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Hands-on Exercises Plan (Integrated Project) for the Course "Design for Electrical and Computer Engineers"

1. Project Overview

In this course, students will work on a **semester-long integrated engineering design project**. The project simulates a **real industrial engineering scenario** and connects theoretical design concepts with practical implementation.

The project scenario is:

Design of a Jet Engine Health Monitoring System for Predictive Maintenance

In this scenario, an airline company experiences unexpected jet engine failures and needs an intelligent monitoring system to detect degradation and predict failures before they occur.
Students will design a **decision-support system for maintenance engineers** that:

- analyzes jet engine sensor data
- detects degradation patterns
- predicts remaining useful life (RUL)
- recommends maintenance actions

The project combines:

- Engineering Design Methodology
- Data Analysis and Visualization
- Machine Learning Prototyping
- System Modeling and Testing
- Reliability Analysis
- Engineering Documentation

The project uses the **NASA C-MAPSS Turbofan Engine Degradation Dataset**, which contains simulated multivariate sensor measurements collected from jet engines operating under different conditions until failure.

All practical work will be performed using:

- **Google Colab**
- **Python**
- **NumPy**
- **Pandas**
- **Matplotlib**
- **Seaborn**

The final outcome will be a **reproducible notebook-based prototype** of a predictive maintenance system.

2. Team Organization

Students will work in **teams of 7 members**.
Each team must:

- create a shared project notebook
- maintain a versioned project repository
- submit weekly progress reports
- prepare group and individual deliverables

Individual accountability will be enforced through:

- individual contribution reports
- peer evaluation checkpoints
- traceable evidence of work (plots, notebooks, code sections)

3. Project Learning Workflow

The project follows the **engineering design lifecycle**, synchronized with the weekly lectures.
The workflow progresses through the following stages:

1. Problem exploration
2. Requirements engineering
3. Concept generation
4. Design evaluation
5. System modeling
6. Prototype development
7. Testing and verification
8. Design iteration and optimization
9. Documentation and communication
10. Ethics and professional responsibility

This structure ensures that students **apply the concepts immediately after they are introduced in lectures**.

4. Weekly Project Milestones

Week 1 – Project Kickoff

Students are introduced to the engineering design problem and dataset.
Tasks:
• Form project teams
• Load the C-MAPSS dataset in Google Colab
• Explore dataset structure
• Plot sensor trends for a sample engine
Deliverables:
• Team formation
• Project notebook initialized

Week 2 – Exploratory Data Analysis

Students analyze the dataset to understand engine degradation behavior.
Tasks:
• Perform exploratory data analysis (EDA)
• Analyze sensor distributions
• Visualize engine trajectories
• Identify missing values and anomalies
Deliverable:
EDA Checkpoint 1
(plots and short insights)

Week 3 – Problem Definition

Students define the engineering problem and stakeholder needs.
Tasks:
• Draft a problem statement
• Identify stakeholders
• Define system needs based on dataset characteristics
Deliverable:
Problem Statement + Needs List (Version 1)

Week 4 – Team Planning

Teams establish project organization.
Tasks:
• Create a team charter
• Define team roles
• Develop a Work Breakdown Structure (WBS)
• Create a preliminary Gantt chart
Deliverable:
Team Charter + Baseline WBS

Week 5 – Requirements Engineering

Students convert needs into engineering requirements.

Tasks:

• Define system requirements
• Specify measurable acceptance criteria
• Map requirements to system functions

Deliverable:

Group Assignment #2 – Requirements Specification (Draft)

Week 6 – Concept Generation

Teams develop alternative system designs.

Example concepts:

• threshold-based monitoring
• machine learning prediction
• hybrid approach

Tasks:

• evaluate feasibility of each concept
• perform quick experiments using sensor data

Deliverable:

Requirements Specification (Final Version)

Week 7 – Design Evaluation

Students apply decision-making methods to select the best design.

Tasks:

• apply AHP or ANP
• evaluate concept alternatives
• analyze trade-offs (accuracy, interpretability, reliability)

Deliverable:

Group Assignment #1 – AHP/ANP

Week 8 – Midterm Design Review

Teams present their progress and receive feedback.

Presentation must include:

• problem definition
• stakeholder needs
• system requirements
• concept selection
• EDA results
• project timeline

Deliverable:

Midterm Project Progress Presentation

Week 9 – System Modeling

Students develop system architecture and behavioral models.
Tasks:
• design system decomposition
• create dataflow diagrams
• model system behavior
Deliverable:
Architecture and Behavioral Models (Version 1)

Week 10 – Prototype Development

Teams build the first working prototype.
Tasks:
• implement baseline predictive model
• create monitoring visualizations
• build preprocessing pipeline
Deliverable:
Start **Group Assignment #4 – Testing Plan**

Week 11 – Design Iteration

Students improve system performance.
Tasks:
• optimize model parameters
• engineer new features
• analyze reliability trade-offs
Deliverable:
Group Assignment #3 – Functional Decomposition and Behavioral Models

Week 12 – Documentation

Teams prepare the structure of the final report.
Tasks:
• organize experiment results
• draft report structure
• design presentation slides
Deliverable:
Report Draft 1 + Slide Draft 1

Week 13 – Ethics and Risk Analysis

Students analyze ethical and societal implications.
Topics include:
• algorithmic bias
• reliability risks
• safety implications of maintenance decisions
Deliverable:
Ethics and Risk Analysis Section

Week 14 – Integration and Final Testing

Teams finalize system implementation.
Tasks:
• integrate all modules
• finalize testing procedures
• rehearse final presentation
Deliverable:
Group Assignment #4 – Testing Report

Week 15 – Final Project Presentation

Each team presents its final system prototype.
Presentation must include:
• system architecture
• analysis results
• model performance
• design trade-offs
• testing results
• ethical considerations
Deliverables:
• Final group presentation
• Final group report
• Project notebook/code

Week 16 – Final Written Exam

Students complete the final written examination.
Students must also submit:
• **Individual contribution report**

5. Final Project Deliverables

Each team must submit:

Group Deliverables

• Final project report
• Google Colab notebook
• project code and visualizations
• presentation slides
• testing documentation

Individual Deliverables

Each student must submit an **individual contribution report** explaining:

• personal responsibilities
• implemented components
• produced figures or code
• testing contributions
• meeting participation
Evidence must be included (notebook sections, figures, logs).

6. References

1. **NASA C-MAPSS Turbofan Engine Degradation Dataset**
   Dataset of simulated turbofan engine sensor measurements recorded until failure.
   https://www.kaggle.com/datasets/behrad3d/nasa-cmaps
2. **NASA C-MAPSS Dataset Description (NASA Dashlink)**
   Official dataset description and documentation.
   https://c3.ndc.nasa.gov/dashlink/resources/14/
3. **Design for Electrical and Computer Engineers: Theory, Concepts, and Practice**
   Ralph M. Ford & Chris S. Coulston
   https://www.amazon.com/Design-Electrical-Computer-Engineers-Ralph/dp/0073380350
4. **Google Colab Documentation**
   Cloud-based Python notebook environment.
   https://developers.google.com/colab
5. **NIST Artificial Intelligence Risk Management Framework (AI RMF 1.0)**
   Framework for managing risks in AI systems.
   https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-ai-rmf-10
6. **ACM Code of Ethics and Professional Conduct**
   https://www.acm.org/about-acm/code-of-ethics
7. **IEEE Code of Ethics**
   https://ieee-cas.org/about/ieee-code-ethics
8. **Quality Function Deployment (QFD / House of Quality)**
   https://www.ifm.eng.cam.ac.uk/research/dmg/tools-and-techniques/quality-function-deployment-qfd/
9. **SuperDecisions Software for AHP/ANP**
   https://www.superdecisions.com
10. **AHPy Python Library**
    https://pypi.org/project/AHPy/

Thank you